

Mechanization of Reasoning :-

* Every X is Y
 A is X

Therefore, A is Y.

* Every politician is clever

Manoj is a politician

Therefore, Manoj is clever.

This is an effort of mechanizing the rules of inference or simply reasoning.

Mechanization of reasoning leads to automated deduction.

Satisfiable :-

A set of formula is called satisfiable if for a set of truth values of the variables in the formula, all the formulas are true.

e.g.:-

(1) $\{P, Q\}$ is satisfiable as both the formula are true when P and Q is true.

(2) $\{P, \neg P\}$ is not satisfiable.

(3) $\{P, \neg P \vee Q\}$ is satisfiable.

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(b) Consistence:-

A set of formula is called consistent if we can not derive a contradiction from the set.

OR

A set of formula is called consistent if there is no formula P such that both P and $\neg P$ can be proved from the given premises and deductive system of formula.

(c) Applications of propositional logic

- (1) Programming Language.
- (2) Software development
- (3) Cryptography and Artificial intelligence.
- (4) Graph theory and digital logic.
- (5) Permutation and combination.
- (6) formal Language etc.

Russell's Paradox :-

In mathematical logic, Russell's paradox is a set-theoretic paradox published by the British philosopher and mathematician Bertrand Russell in 1901.

Russell's paradox shows that every set theory that contains an unrestricted comprehension principle leads to contradictions.

Russell's paradox (R) is a set of all sets that are not members of themselves.

$$R = \{x : x \notin x\}, \text{ then } R \in R \Leftrightarrow R \notin R$$

Now, consider two cases :

- (1) If, $R \in R$, then set R contains itself. $\rightarrow$ Contradiction.
- (2) If, $R \notin R$, then set R does not contain itself, but according to the definition, R must contain all sets that do not contain themselves. $\rightarrow$ Contradiction.

This is a paradox.

[Barber Paradox]

e.g.:- There is a city x , where barber does the shave for all those men in the city who do not shave themselves. Now, the question is who does the shave for the barber.

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[God Paradox]

Can God create a stone that he can not lift up?

This is based on the general assumption that everything is possible to god. so if he creates a stone which he can not lift up, then it breaks the condition he can lift anything. At the same time if he can not create a stone, then its against that he can create anything.